

EDA
CAPSTONE PROJECT
HOTEL BOOKING ANALYSIS
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Problem Statement:

1. How do booking cancellation patterns differ between City Hotel and Resort Hotel?
2. How do booking patterns vary across months for City Hotels and Resort Hotels?
3. How are successful hotel bookings distributed across different countries worldwide?
4. Which market segments contribute the most bookings for each hotel type?
5. How does the Average Daily Rate (ADR) vary across months for City Hotel and Resort Hotel?
6. Which deposit policies are associated with the highest booking cancellation rates?
7. Which customer types have the highest cancellation rates?

Dataset Overview:

The dataset contains hotel reservation information for two hotel categories: City Hotels and Resort Hotels.

A total of 119,390 booking records and 32 features are included in the dataset. It contains approximately 79,330 City Hotel bookings and 40,060 Resort Hotel bookings.

The dataset includes both cancelled and successful bookings, making it suitable for analyzing booking patterns, cancellation behavior, customer segments, pricing trends, and geographical demand.

After data cleaning and preprocessing, 87,228 records remained for analysis.

Cleaning Data

The quality of data plays an important role in Exploratory Data Analysis (EDA). Data cleaning was performed to remove inconsistencies and improve the reliability of the analysis.

The following steps were carried out during the data cleaning process:

1. Removed duplicate records from the dataset.
2. Handled missing values in relevant columns.
3. Removed invalid bookings where the total number of guests (adults, children, and babies) was zero.
4. Removed records containing undefined market segments.
5. Verified data consistency and prepared the dataset for further analysis.

Data Summary

Hotel – Indicates the type of hotel (City Hotel or Resort Hotel).

Is_Canceled – Shows whether a booking was cancelled or not. Values: 0 = Not Cancelled, 1 = Cancelled.

Arrival_Date_Month – Month in which the customer is scheduled to arrive.

Country – Country of origin of the guest.

Market_Segment – Indicates the booking source or purpose of travel (e.g., Online TA, Offline TA/TO, Corporate, Direct).

Distribution_Channel – Channel through which the booking was made.

Deposit_Type – Type of deposit associated with the reservation
(No Deposit, Non Refund, Refundable).

Customer_Type – Classification of customers such as Transient,
Contract, Group, and Transient-Party.

ADR (Average Daily Rate) – Average revenue earned per occupied room per
day.

Reservation_Status – Final status of the reservation
(Check-Out, Canceled, or No-Show).

Libraries Used

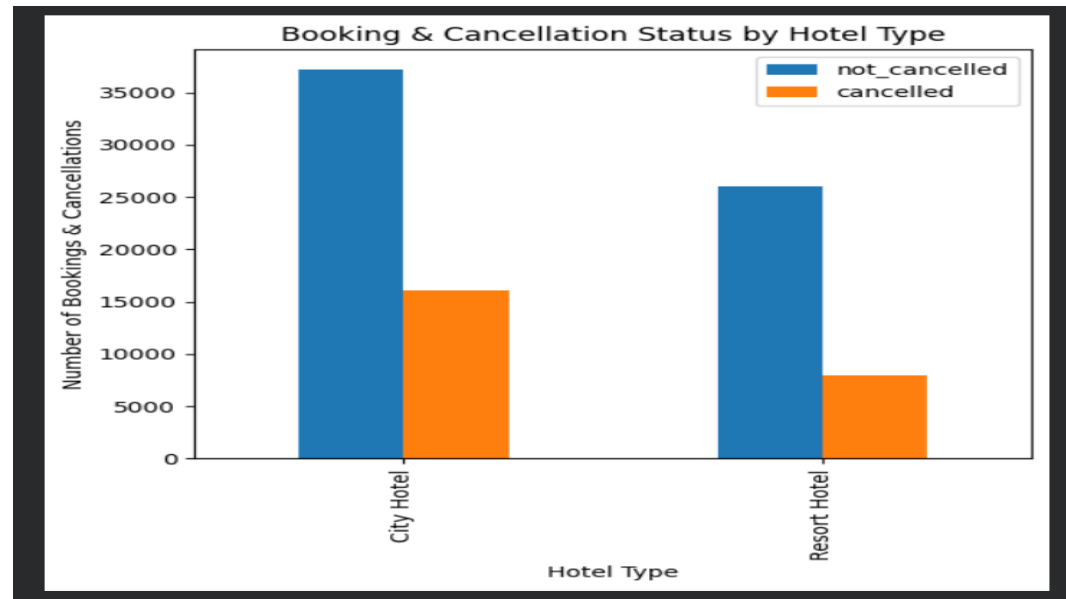
- ❖ import numpy as np
- ❖ import pandas as pd
- ❖ import matplotlib.pyplot as plt
- ❖ import seaborn as sns
- ❖ import plotly.express as px
- ❖ import folium



Exploratory Data Analysis (EDA)

1. How do booking cancellation patterns differ between City Hotel and Resort Hotel?

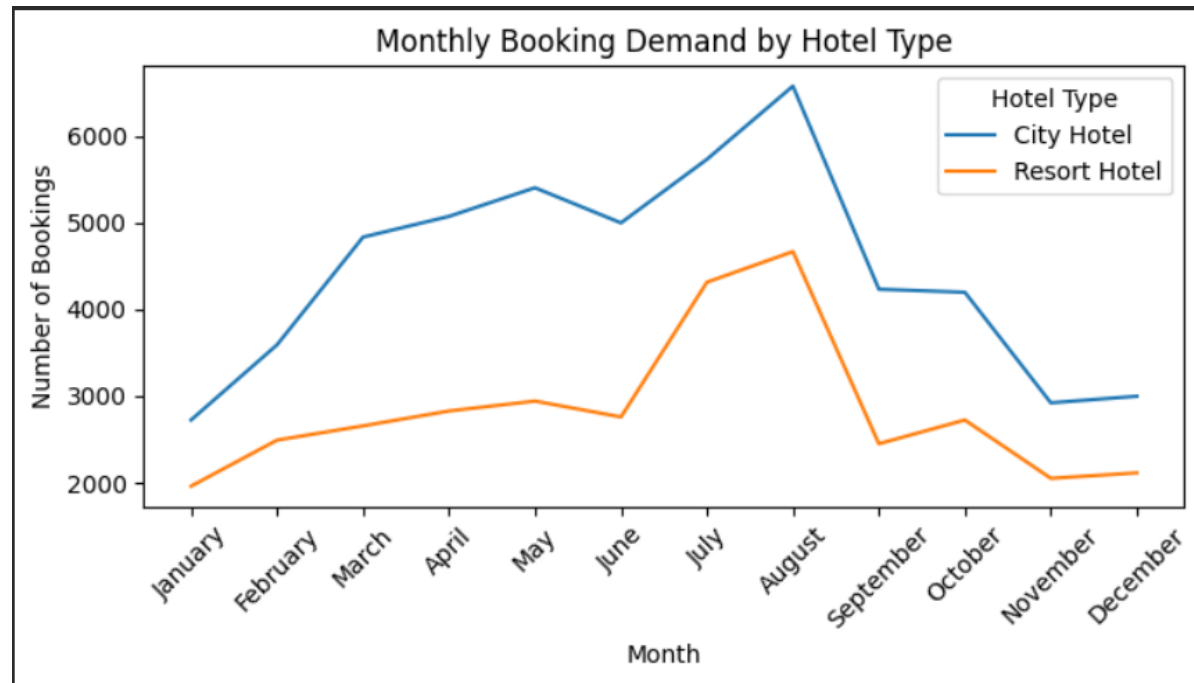
	hotel	not_cancelled	cancelled
0	City Hotel	37239	16033
1	Resort Hotel	25982	7974



- City Hotels have a higher number of bookings and cancellations than Resort Hotels.

How do booking patterns vary across months for City Hotel and Resort Hotel?

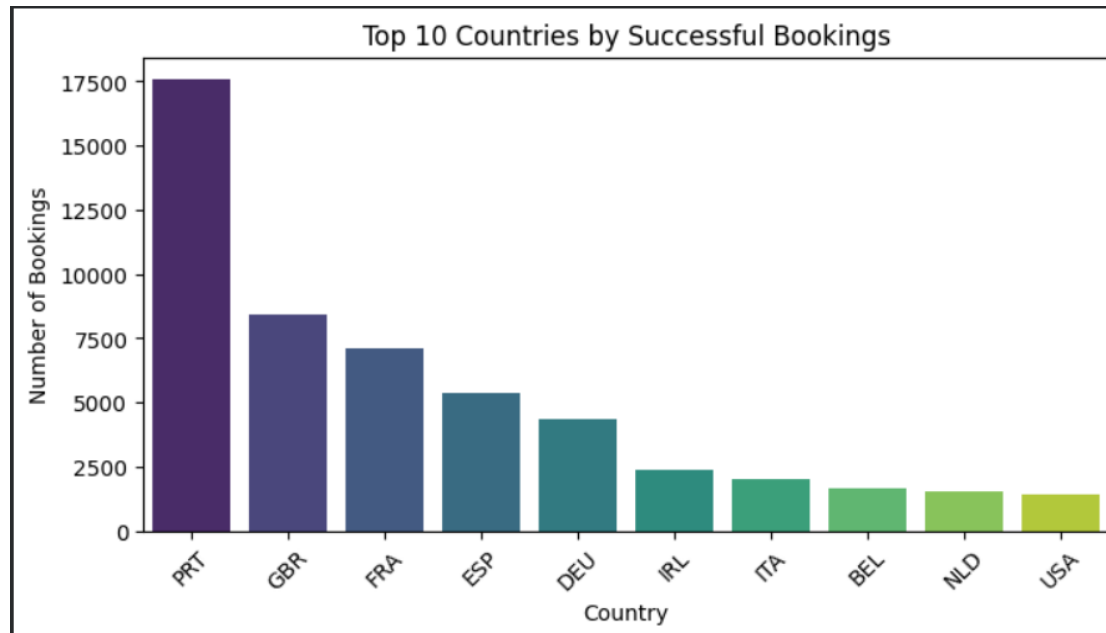
monthly_hotel			
hotel	month	City Hotel	Resort Hotel
0	January	2724	1961
1	February	3591	2492
2	March	4833	2656
3	April	5072	2828
4	May	5402	2942
5	June	4997	2759
6	July	5730	4313
7	August	6574	4666
8	September	4232	2450
9	October	4197	2724
10	November	2922	2051
11	December	2998	2114



- Booking demand peaks during July and August for both hotel types, while the lowest demand is observed during November, December, and January.

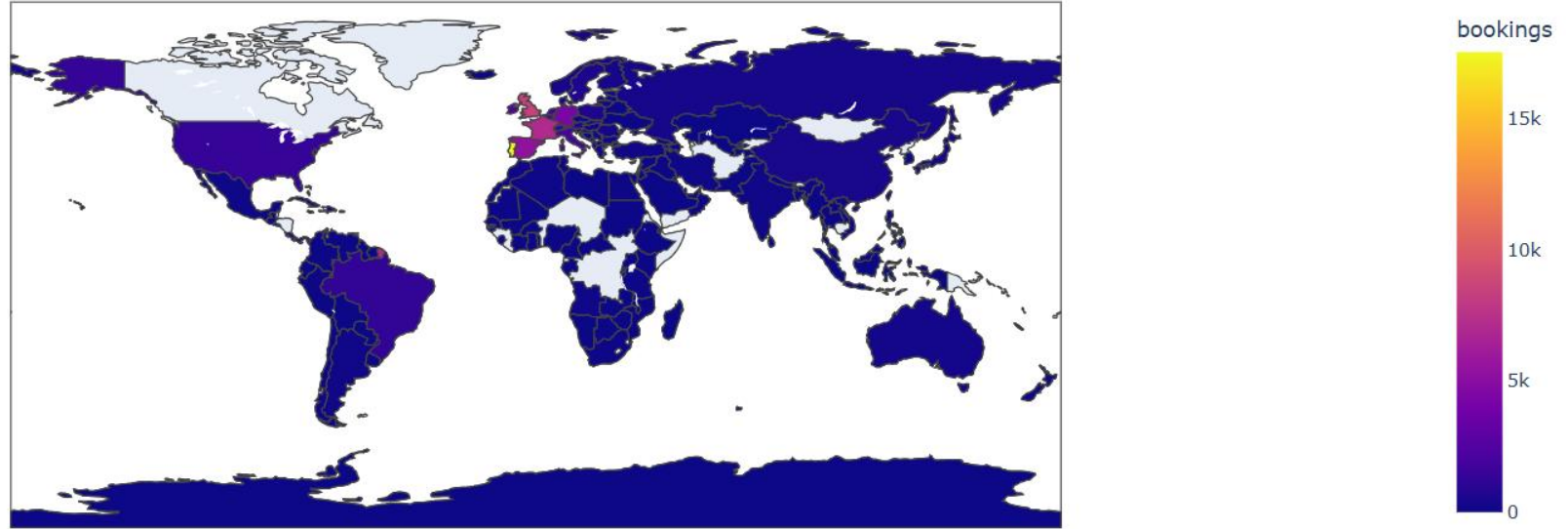
How are successful hotel bookings distributed across different countries worldwide?

	country	bookings
0	PRT	17573
1	GBR	8440
2	FRA	7091
3	ESP	5382
4	DEU	4332
5	IRL	2347
6	ITA	1986
7	BEL	1670
8	NLD	1560
9	USA	1412



- Successful bookings are heavily concentrated in Portugal, with the United Kingdom, France, Spain, and Germany representing other major source markets.

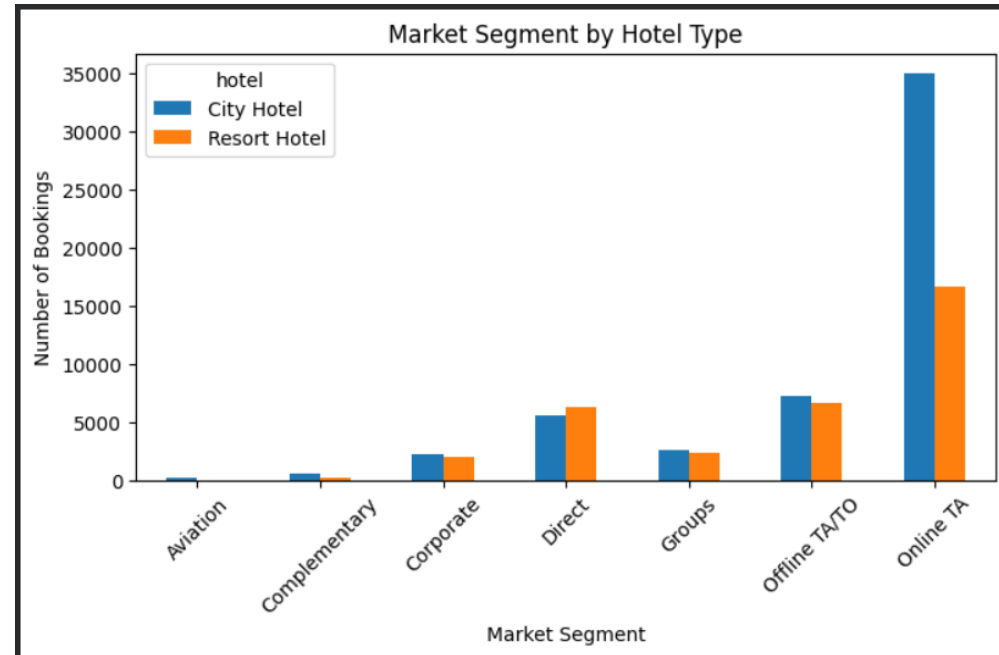
Successful Hotel Bookings by Country



- The majority of successful bookings originate from European countries, particularly Portugal, the United Kingdom, France, and Spain.

Which market segments contribute the most bookings for each hotel type?

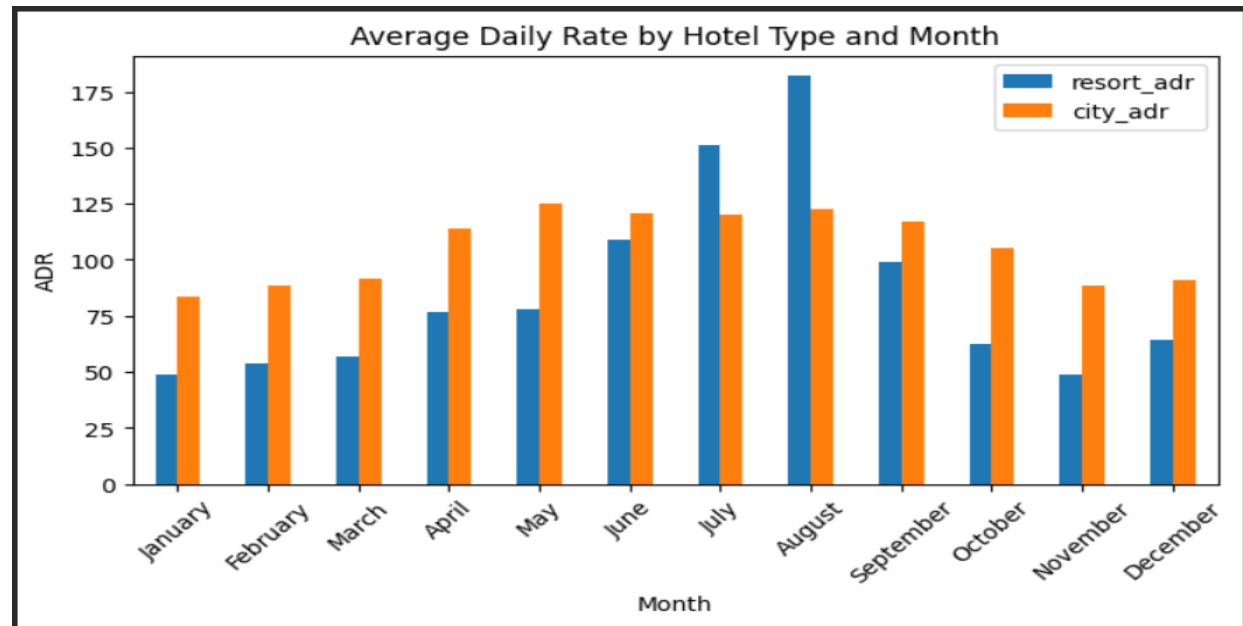
hotel	City Hotel	Resort Hotel
market_segment		
Aviation	226	0
Complementary	503	189
Corporate	2218	1982
Direct	5538	6242
Groups	2619	2303
Offline TA/TO	7239	6616
Online TA	34929	16624



- Online Travel Agencies (Online TA) generate the highest number of bookings for both hotel types, making them the most important distribution source.

How does the Average Daily Rate (ADR) vary across months for City Hotel and Resort Hotel?

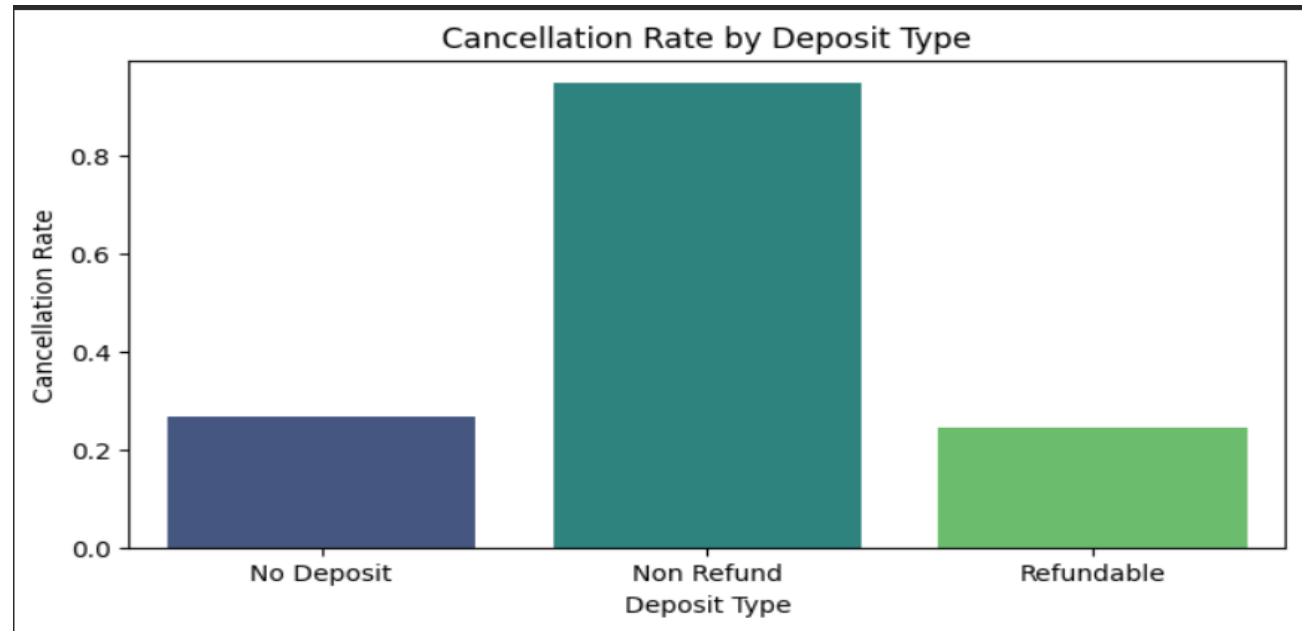
	month	resort_adr	city_adr
0	January	48.599615	83.378062
1	February	53.624311	88.210810
2	March	57.110018	91.306933
3	April	76.841664	114.251636
4	May	77.725129	124.825944
5	June	109.026942	120.673753
6	July	150.914540	120.069489
7	August	182.103524	122.825388
8	September	98.855456	116.973279
9	October	62.634853	105.517410
10	November	48.956548	88.271089
11	December	64.072826	90.769559



- ADR increases significantly during the summer months, with Resort Hotels recording the highest ADR in July and August.

Which deposit policies are associated with the highest booking cancellation rates?

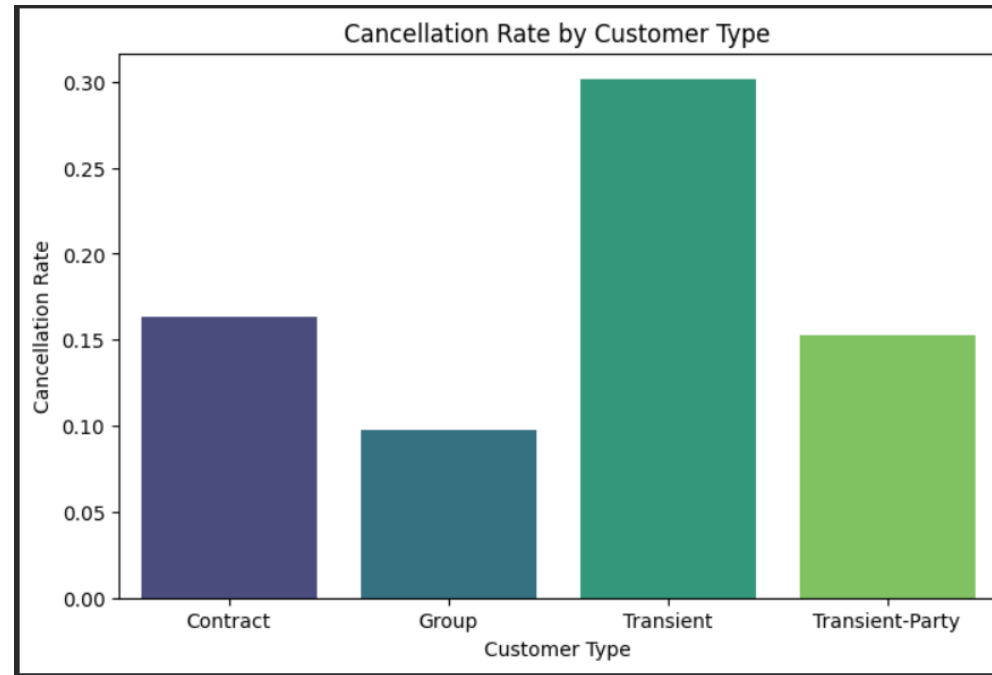
	deposit_type	cancel_rate
0	No Deposit	0.267161
1	Non Refund	0.947013
2	Refundable	0.242991



- Non-Refund bookings have the highest cancellation rate, significantly exceeding No Deposit and Refundable bookings.

Which customer types have the highest cancellation rates?

	customer_type	cancellations
0	Contract	0.163317
1	Group	0.097967
2	Transient	0.301425
3	Transient-Party	0.152352



- Transient customers are the most likely to cancel bookings, making them a key segment for cancellation reduction strategies.

Conclusion

- City Hotels receive the highest number of bookings and cancellations.
- Booking demand peaks during July and August, indicating strong seasonal trends.
- Online TA is the most important booking channel for both hotel types.
- Transient customers and Non-Refund bookings show the highest cancellation rates.
- ADR increases during peak travel months, especially for Resort Hotels.
- Portugal, the United Kingdom, France, and Spain are the major source markets for successful bookings.
- Hotels can improve revenue by optimizing pricing, reducing cancellations, and focusing on high-performing customer segments and booking channels.